CHAPTER FOUR

SYSTEM DESIGN

4.1 Overview

System design is the stage in which the system requirements and analysis models are transformed into a detailed technical plan for system implementation. This chapter presents the design of the **UniMap, Smart University Map and Navigation System**, describing how the system components are organized, how they interact, and how data flows within the system.

The design focuses on creating a scalable, maintainable, and user-friendly system that meets both functional and non-functional requirements. The UniMap system is designed using modern web and mobile technologies and follows a modular architecture that separates the user interface, application logic, and data storage. This approach improves system reliability and allows future expansion and maintenance.

4.2 System Design

The UniMap system is designed as a **client–server application** that supports both web and mobile platforms. Users interact with the system through a web-based interface, while backend services handle business logic, routing operations, and data management.

The system design emphasizes:

- Modularity and separation of concerns
- Ease of maintenance and scalability
- Efficient data processing and secure access

This design approach ensures that the system can serve a large number of users while maintaining acceptable performance.

4.3 Architecture of the System

The UniMap system follows a **layered architecture**, which consists of the following major components:

1. Presentation Layer

- Provides the graphical user interface for users.
- Displays the interactive campus map, search functionality, and route visualization.

- Designed to be responsive and accessible on both desktop and mobile devices.

2. Application Layer

- Contains the core business logic of the system.
- Processes user requests such as location search and route generation.
- Manages authentication and authorization for administrative users.
- Communicates with the data layer through well-defined interfaces.

3. Data Layer

- Stores all system data, including locations, routes, facilities, events, and user information.
- Ensures data integrity, consistency, and secure storage.

4. Mobile Deployment Layer

- Enables the system to be packaged as a mobile application.
- Allows deployment to mobile platforms without rewriting the core application logic.

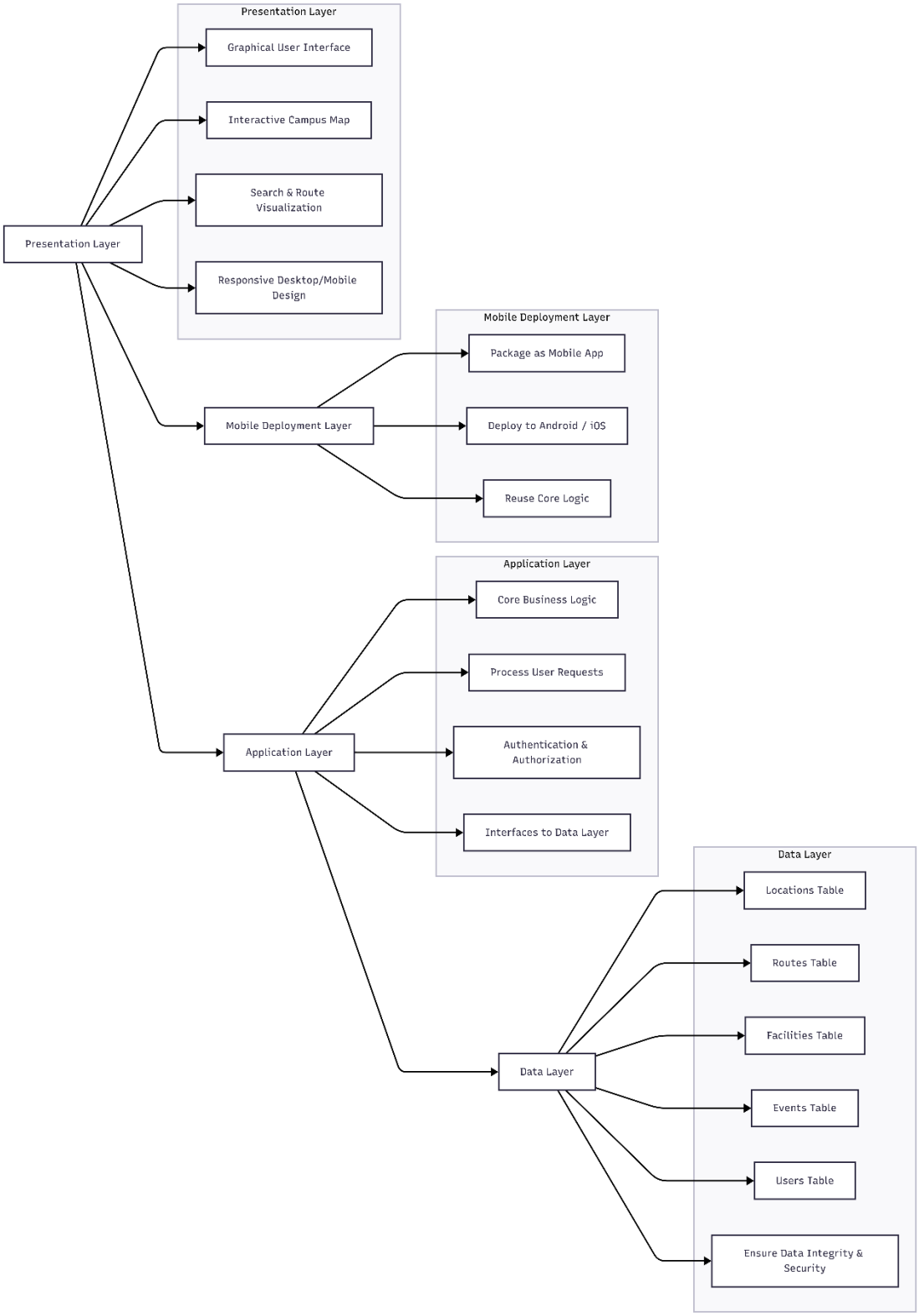

Figure 4.1: Architecture Diagram of the UniMap System

4.4 Database Design

The database design defines how system data is organized, stored, and managed. The UniMap system uses a structured database to store information related to campus locations, navigation routes, facilities, events, and user roles.

The database is designed to:

- Support efficient querying and data retrieval
- Maintain data integrity through relationships and constraints
- Allow easy updates and future expansion

Key database entities include:

- User
- Location
- Route
- Facility
- Event

Relationships between these entities are established to reflect real-world associations within the university campus.

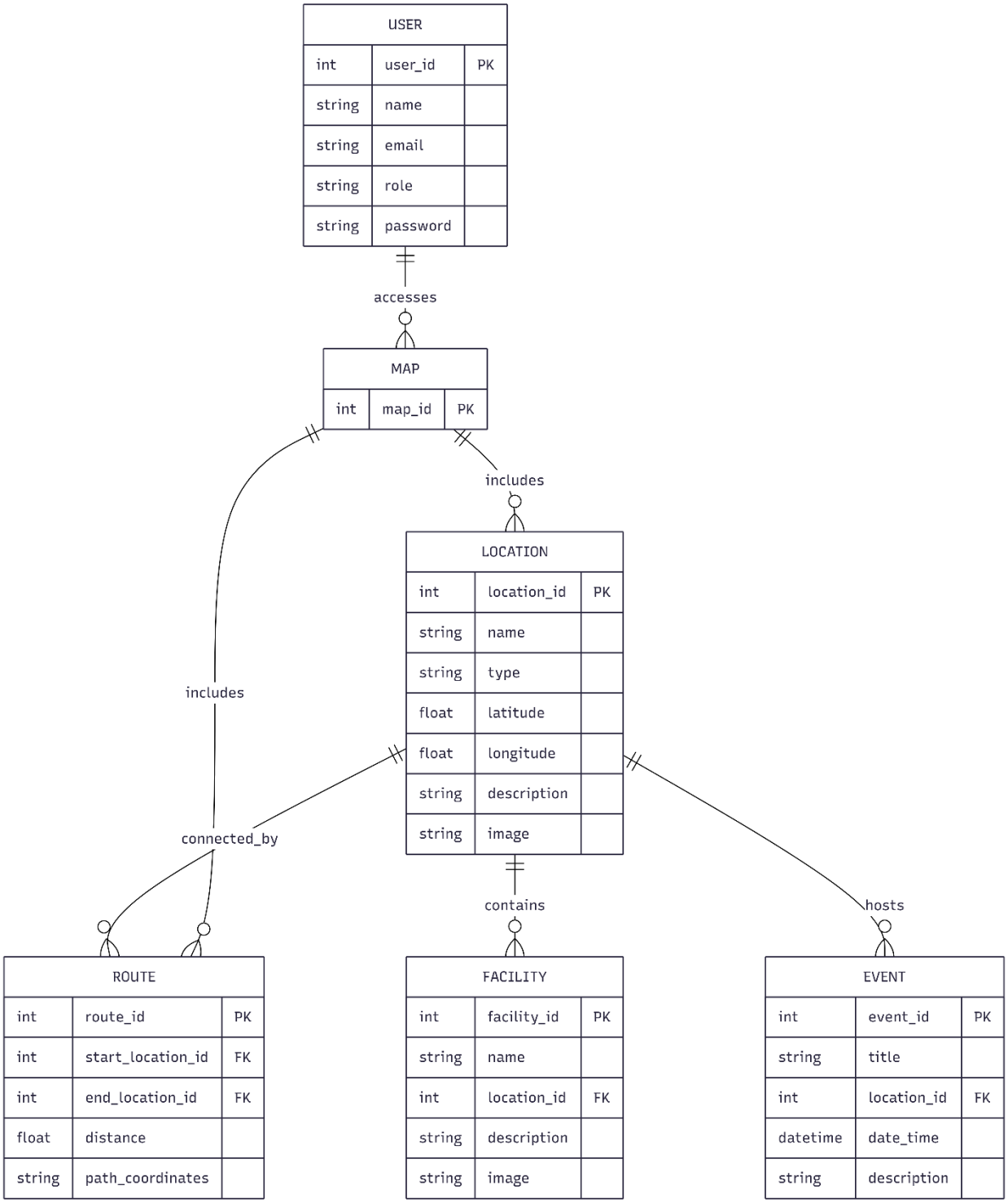

Figure 4.2: ER Diagram of the UniMap System Database

4.5 User Interface Design

The user interface design focuses on usability, simplicity, and accessibility. The system provides a clean and intuitive interface that allows users to easily navigate the campus and access required information.

Key interface components include:

- Home page displaying the campus map

Figure 4.3: UI Mockups Home page

- Jigjiga University Map

Figure 4.4: UI Mockups The University Map

- Search bar for buildings, offices, and departments

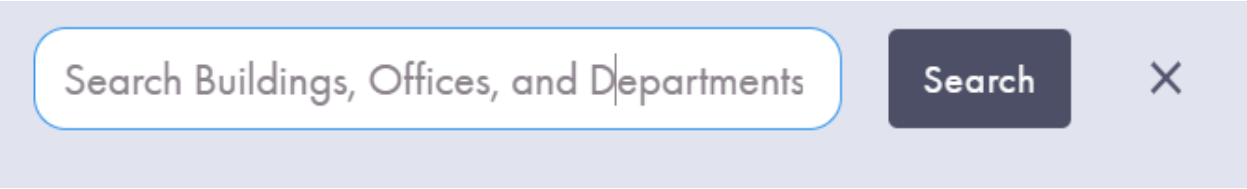

Figure 4.5: UI Mockups Home Search Bar

- Route visualization with clear indicators

Figure 4.6: UI Mockups Route

- Information panels for facilities and events
- Administrative dashboard for data management

The interface is designed to be responsive, ensuring consistent user experience across different screen sizes and devices.